Meta-Analysis_TMS_Physical_Activity

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R Markdown file set-up and packages

Packages required: 1) dmetar 2) tidyverse 3) meta 4) metafor 5) esc 6) TOSTER

Meta-analysis

Import Pearson r data

Data file glimpse

Calculating z and var.z

Excitability (CSE): Primary analysis of using metafor::rma.mv()

```
m.cor.excitability <- rma.mv(yi = z, # pool correlation coefficients Fishers Z transformation https://b
                V = var.z, # sampling variance of Fisher's Z values: 1/(N-3), where N = the sample size
                slab = author,
                data = excitability_r,
                random = ~ 1 | author/cor_id,
                test = "t",
                method = "REML")
summary(m.cor.excitability)
```

```
##
## Multivariate Meta-Analysis Model (k = 27; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##   4.4134   -8.8269   -2.8269    0.9474   -1.7360
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0000  0.0000     18     no         author
## sigma^2.2  0.0000  0.0000     27     no  author/cor_id
##
## Test for Heterogeneity:
## Q(df = 26) = 26.0827, p-val = 0.4586
##
## Model Results:
##
## estimate      se     tval  df    pval    ci.lb   ci.ub
##  -0.0259  0.0382  -0.6775  26  0.5041  -0.1044  0.0527
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Excitability (CSE): Converting Fisher's z back to Pearson's r

```
round(convert_z2r(-0.0287), 2) # point estimate
```

```
## [1] -0.03
```

```r
round(convert_z2r(-0.1190), 2) # lower CI
```

```
## [1] -0.12
```

```r
round(convert_z2r(0.0617), 2) # Upper CI
```

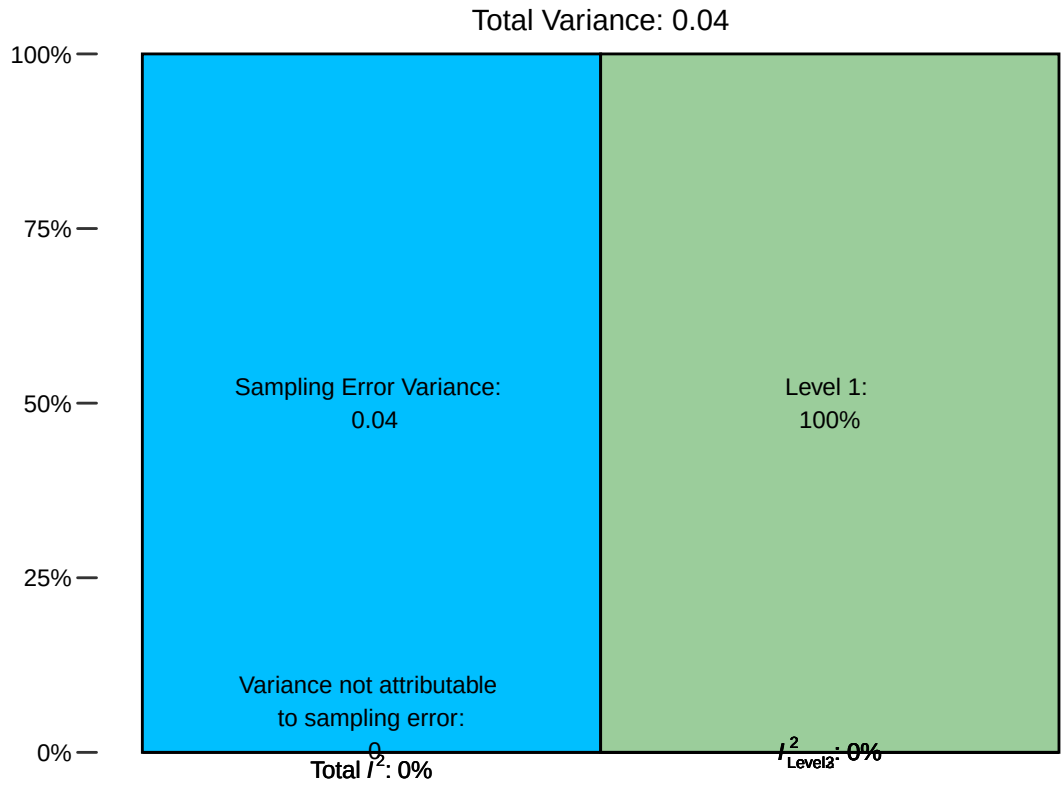
```
## [1] 0.06
```

Excitability (CSE): Explore heterogeneity

```r
i2 <- var.comp(m.cor.excitability)
summary(i2)
```

```
##         % of total variance  I2
## Level 1        1.000000e+02 ---
## Level 2        4.812659e-08   0
## Level 3        5.857891e-09   0
## Total I2: 0%
```

```r
plot(i2)
```



ADD TEXT

Excitability (CSE): Sensitivity analysis using meta::metacor()

```
m.excitability <- metacor(cor = cor,
              n = n,
              studlab = author,
              data = excitability_r,
              cluster = cluster,
              common = FALSE,
              random = TRUE,
              method.tau = "REML",
              method.random.ci = "HK",
              title = "Sensitivity meta-analysis using metacor")

m.excitability
```

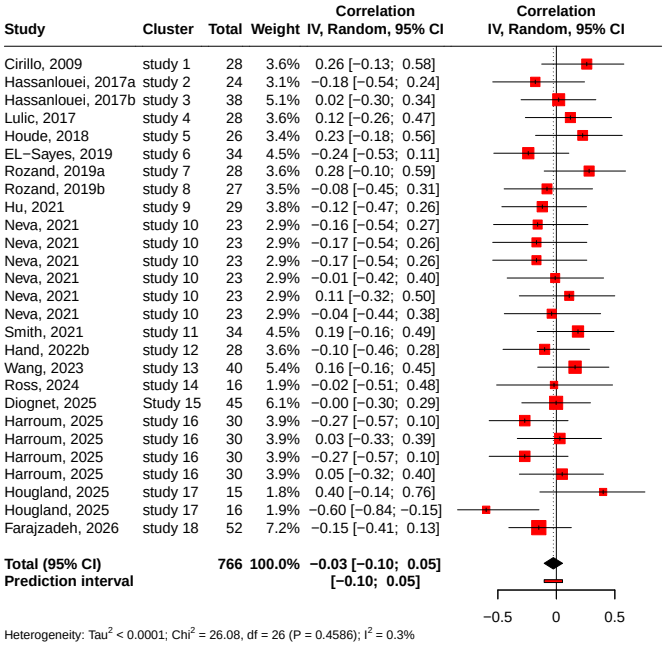
```
## Review:     Sensitivity meta-analysis using metacor
##
## Number of studies: n = 18
## Number of estimates: k = 27
## Number of observations: o = 766
##
##                           COR            95%-CI     t p-value
## Random effects model -0.0259 [-0.1040; 0.0526] -0.68  0.5041
##
## Quantifying heterogeneity (with 95%-CIs):
##  tau^2.1 < 0.0001 [0.0000; 0.0236]; tau.1 < 0.0001 [0.0000; 0.1537] (between cluster)
##  tau^2.2 < 0.0001 [0.0000; 0.0280]; tau.2 < 0.0001 [0.0000; 0.1675] (within cluster)
##  I^2 = 0.3% [0.0%; 42.7%]; H = 1.00 [1.00; 1.32]
##
## Test of heterogeneity:
##      Q d.f. p-value
##  26.08   26  0.4586
##
## Details of meta-analysis methods:
## - Inverse variance method (three-level model)
## - Restricted maximum-likelihood estimator for tau^2
## - Profile-Likelihood method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Random effects confidence interval based on t-distribution (df = 26)
## - Fisher's z transformation of correlations
```

#Corticospinal excitability ADD TEXT

Excitability (CSE): Forest plot

```
forest(m.excitability,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
```

```
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)
```



Excitability (CSE): Save plot

```
library(svglite)

svglite("forest_plot_excitability.svg", height = 7, width = 7)

forest(m.excitability,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)

dev.off()
```

```
## pdf
##   2
```

Inhibition (SICI, LICI, cSP, iSP): Primary analysis of using metafor::rma.mv()

```
m.cor.inhibition <- rma.mv(yi = z, # pool correlation coefficients Fishers Z transformation https://boo
                V = var.z, # sampling variance of Fisher's Z values: 1/(N-3), where N = the sample size
                slab = author,
                data = inhibition_r,
                random = ~ 1 | author/cor_id,
                test = "t",
                method = "REML")
summary(m.cor.inhibition)
```

```
##
## Multivariate Meta-Analysis Model (k = 20; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##   1.5663   -3.1325    2.8675    5.7008    4.4675
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0181  0.1346     15     no         author
## sigma^2.2  0.0000  0.0000     20     no  author/cor_id
##
## Test for Heterogeneity:
## Q(df = 19) = 27.6145, p-val = 0.0911
##
## Model Results:
##
## estimate      se     tval  df    pval    ci.lb   ci.ub
##  -0.0011  0.0548  -0.0208  19  0.9836  -0.1158  0.1135
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Inhibition (SICI, LICI, cSP, iSP): Converting Fisher's z back to Pearson's r

```
round(convert_z2r(0.0237), 2) # point estimate
```

```
## [1] 0.02
```

```
round(convert_z2r(-0.1021), 2) # lower CI
```

```
## [1] -0.1
```

```
round(convert_z2r(0.1496), 2) # Upper CI
```

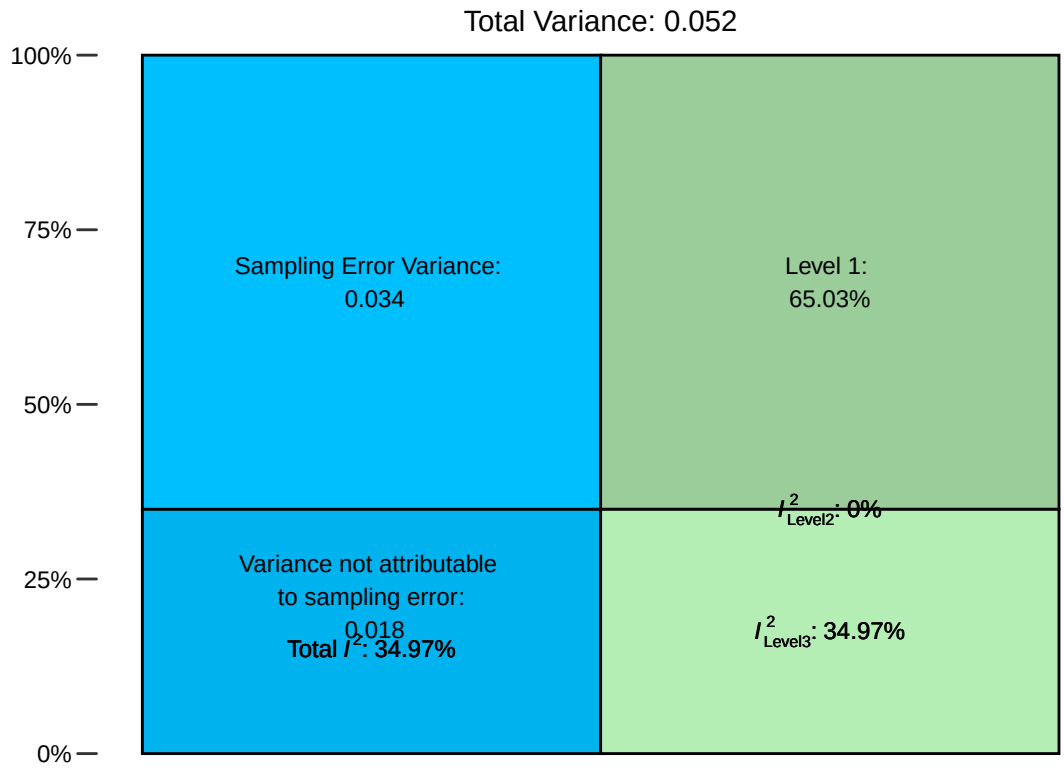
```
## [1] 0.15
```

Inhibition (SICI, LICI, cSP, iSP): Explore heterogeneity

```
i2 <- var.comp(m.cor.inhibition)
summary(i2)
```

```
##         % of total variance    I2
## Level 1        6.502902e+01   ---
## Level 2        3.678491e-08     0
## Level 3        3.497098e+01 34.97
## Total I2: 34.97%
```

```
plot(i2)
```



ADD TEXT

Inhibition (SICI, LICI, cSP, iSP): Sensitivity analysis using meta::metacor()

```
m.inhibition <- metacor(cor = cor,
              n = n,
              studlab = author,
              data = inhibition_r,
              cluster = cluster,
```

```
              common = FALSE,
              random = TRUE,
              method.tau = "REML",
              method.random.ci = "HK",
              title = "Sensitivity meta-analysis using metacor")

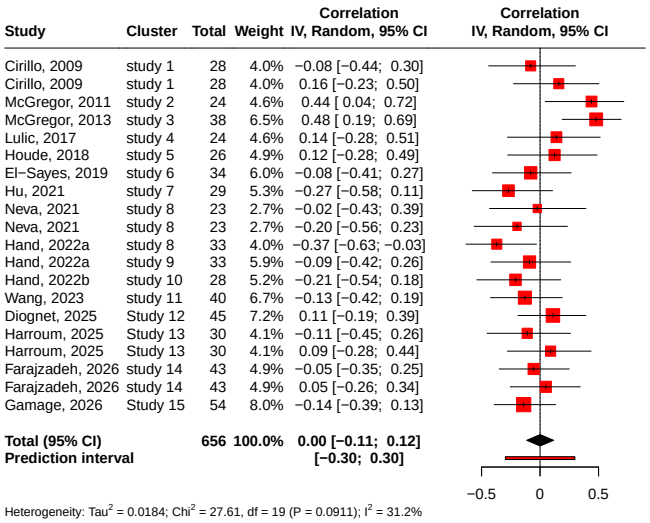
m.inhibition
```

```
## Review:     Sensitivity meta-analysis using metacor
##
## Number of studies: n = 15
## Number of estimates: k = 20
## Number of observations: o = 656
##
##                          COR            95%-CI    t p-value
## Random effects model 0.0008 [-0.1140; 0.1155] 0.01  0.9888
##
## Quantifying heterogeneity (with 95%-CIs):
##  tau^2.1 = 0.0184 [0.0000; 0.0842]; tau.1 = 0.1356 [0.0000; 0.2901] (between cluster)
##  tau^2.2 < 0.0001 [0.0000; 0.0617]; tau.2 < 0.0001 [0.0000; 0.2484] (within cluster)
##  I^2 = 31.2% [0.0%; 60.0%]; H = 1.21 [1.00; 1.58]
##
## Test of heterogeneity:
##      Q d.f. p-value
##  27.61   19  0.0911
##
## Details of meta-analysis methods:
## - Inverse variance method (three-level model)
## - Restricted maximum-likelihood estimator for tau^2
## - Profile-Likelihood method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Random effects confidence interval based on t-distribution (df = 19)
## - Fisher's z transformation of correlations
```

#Cortical inhibition ADD TEXT

Inhibition (SICI, LICI, cSP, iSP): Forest plot

```
forest(m.inhibition,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)
```



Inhibition (SICI, LICI, cSP, iSP): Save plot

```
library(svglite)

svglite("forest_plot_inhibition.svg", height = 7, width = 7)

forest(m.inhibition,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)

dev.off()
```

```
## pdf
##   2
```

Facilitation (ICF, SICF): Primary analysis of using metafor::rma.mv()

```
m.cor.facilitation <- rma.mv(yi = z, # pool correlation coefficients Fishers Z transformation https://b
                V = var.z, # sampling variance of Fisher's Z values: 1/(N-3), where N = the sample size
                slab = author,
                data = facilitation_r,
                random = ~ 1 | author/cor_id,
                test = "t",
                method = "REML")
summary(m.cor.facilitation)
```

```
##
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##   2.8964   -5.7927    0.2073    1.9021    2.8740
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0000  0.0000      8     no         author
## sigma^2.2  0.0000  0.0000     14     no  author/cor_id
##
## Test for Heterogeneity:
## Q(df = 13) = 12.4067, p-val = 0.4946
##
## Model Results:
##
## estimate      se    tval  df    pval    ci.lb   ci.ub
##   0.0600  0.0520  1.1540  13  0.2693  -0.0523  0.1723
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Facilitation (ICF, SICF): Converting Fisher's z back to Pearson's r

```
round(convert_z2r(0.0674), 2) # point estimate
```

```
## [1] 0.07
```

```
round(convert_z2r(-0.0449), 2) # lower CI
```

```
## [1] -0.04
```

```
round(convert_z2r(0.1797), 2) # Upper CI
```

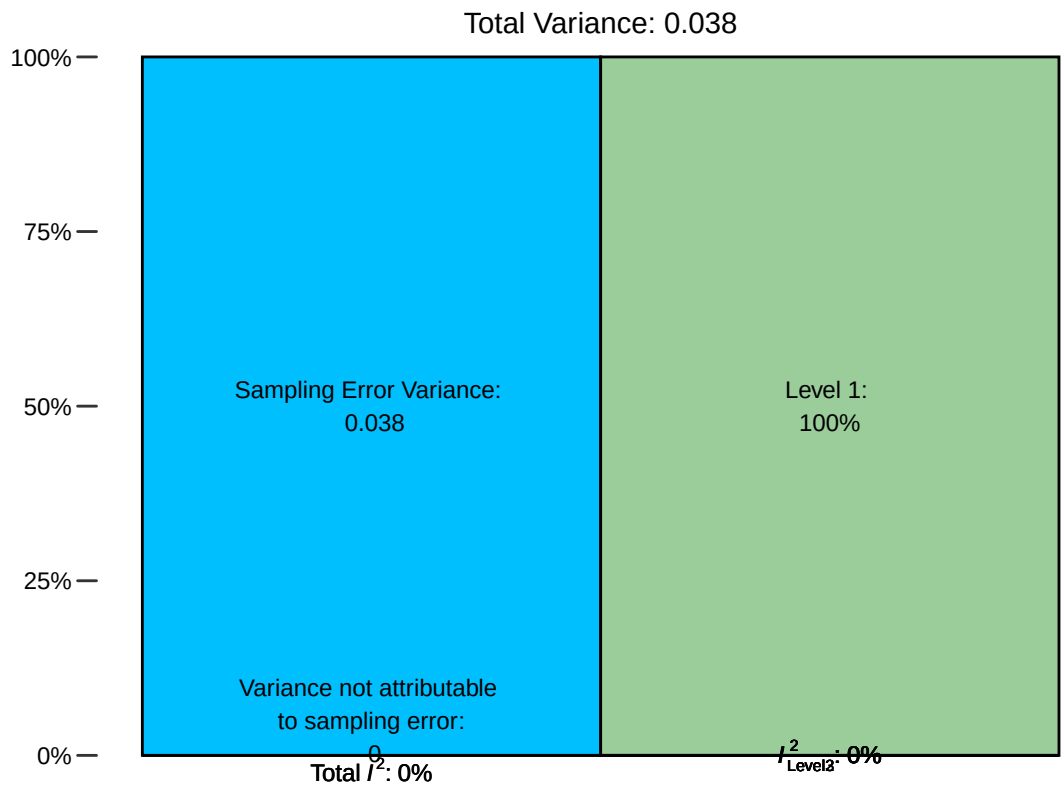
```
## [1] 0.18
```

Facilitation (ICF, SICF): Explore heterogeneity

```
i2 <- var.comp(m.cor.facilitation)
summary(i2)
```

```
##         % of total variance  I2
## Level 1        1.000000e+02 ---
## Level 2        2.762553e-08   0
## Level 3        0.000000e+00   0
## Total I2: 0%
```

```
plot(i2)
```



ADD TEXT

Facilitation (ICF, SICF): Sensitivity analysis using meta::metacor()

```
m.facilitation <- metacor(cor = cor,
              n = n,
              studlab = author,
              data = facilitation_r,
              cluster = cluster,
              common = FALSE,
              random = TRUE,
              method.tau = "REML",
              method.random.ci = "HK",
              title = "Sensitivity meta-analysis using metacor")

m.facilitation
```

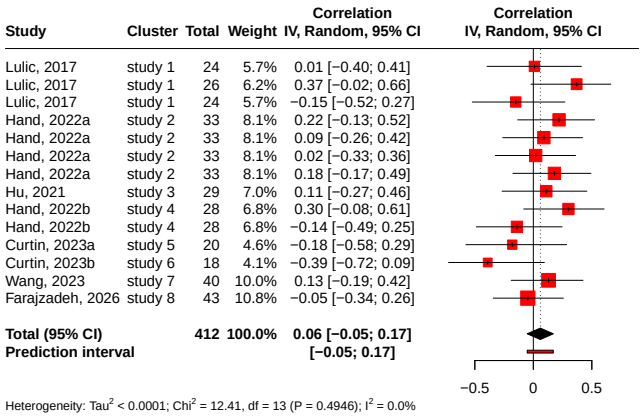
```
## Review:     Sensitivity meta-analysis using metacor
##
## Number of studies: n = 8
## Number of estimates: k = 14
```

```
## Number of observations: o = 412
##
##                          COR            95%-CI    t p-value
## Random effects model 0.0599 [-0.0523; 0.1706] 1.15  0.2693
##
## Quantifying heterogeneity (with 95%-CIs):
##  tau^2.1 = 0      [0.0000; 0.0410]; tau.1 = 0      [0.0000; 0.2025] (between cluster)
##  tau^2.2 < 0.0001 [0.0000; 0.0439]; tau.2 < 0.0001 [0.0000; 0.2095] (within cluster)
##  I^2 = 0.0% [0.0%; 55.0%]; H = 1.00 [1.00; 1.49]
##
## Test of heterogeneity:
##      Q d.f. p-value
##  12.41   13  0.4946
##
## Details of meta-analysis methods:
## - Inverse variance method (three-level model)
## - Restricted maximum-likelihood estimator for tau^2
## - Profile-Likelihood method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Random effects confidence interval based on t-distribution (df = 13)
## - Fisher's z transformation of correlations
```

#Cortical Facilitation ADD TEXT

Facilitation (ICF, SICF): Forest plot

```
forest(m.facilitation,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)
```



Facilitation (ICF, SICF): Save plot

```
library(svglite)

svglite("forest_plot_facilitation.svg", height = 7, width = 7)

forest(m.facilitation,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)

dev.off()
```

```
## pdf
##   2
```

Meta-analysis: Subgroup analyses

Inhibition measure

```
inhibition_r$inhibitory.measure <- as.factor(inhibition_r$inhibitory.measure)

inhibition_tms_subg <- rma.mv(yi = z,
                      V = var.z,
                      slab = author,
                      data = inhibition_r,
```

```
                      random = ~ 1 | author/cor_id,
                      test = "t",
                      method = "REML",
                      mods = ~ inhibitory.measure)

summary(inhibition_tms_subg)
```

```
##
## Multivariate Meta-Analysis Model (k = 20; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##   7.9468  -15.8937   -3.8937    0.7419    5.4397
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0000  0.0000     15     no         author
## sigma^2.2  0.0000  0.0000     20     no  author/cor_id
##
## Test for Residual Heterogeneity:
## QE(df = 16) = 8.2678, p-val = 0.9405
##
## Test of Moderators (coefficients 2:4):
## F(df1 = 3, df2 = 16) = 6.4489, p-val = 0.0045
##
## Model Results:
##
##                          estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt                    0.1438  0.1443   0.9961  16  0.3340  -0.1622  0.4498
## inhibitory.measureiSP      0.3602  0.1967   1.8311  16  0.0858  -0.0568  0.7772
## inhibitory.measureLICI    -0.0947  0.2141  -0.4425  16  0.6640  -0.5486  0.3591
## inhibitory.measureSICI    -0.2390  0.1518  -1.5741  16  0.1350  -0.5608  0.0829
##
## intrcpt
## inhibitory.measureiSP   .
## inhibitory.measureLICI
## inhibitory.measureSICI
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

ADD TEXT

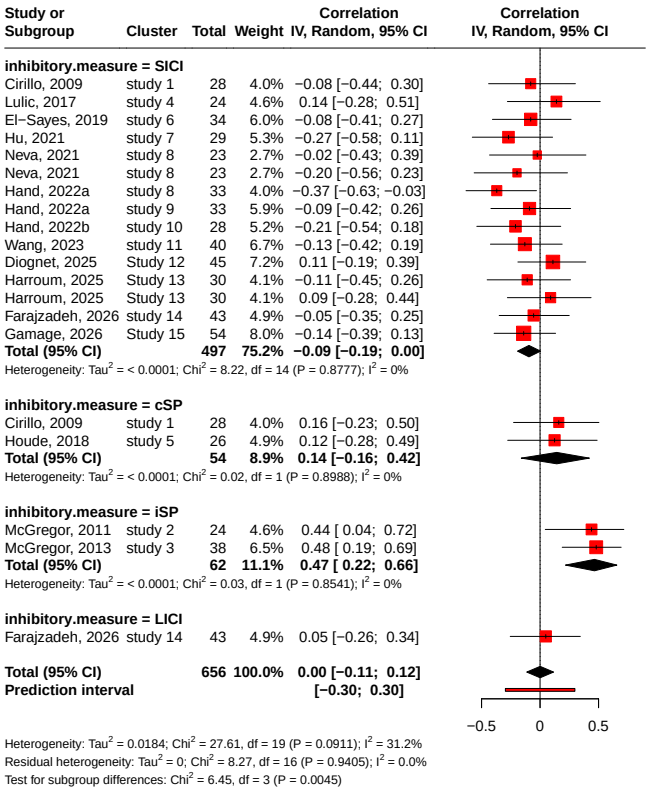
```
inhibition_measure_type <- update(m.inhibition, subgroup = inhibitory.measure, tau.common = TRUE)
inhibition_measure_type
```

```
## Review:     Sensitivity meta-analysis using metacor
##
## Number of studies: n = 15
## Number of estimates: k = 20
## Number of observations: o = 656
##
```

```
##                          COR            95%-CI    t p-value
## Random effects model 0.0008 [-0.1140; 0.1155] 0.01  0.9888
##
## Quantifying heterogeneity (with 95%-CIs):
##  tau^2.1 = 0.0184 [0.0000; 0.0842]; tau.1 = 0.1356 [0.0000; 0.2901] (between cluster)
##  tau^2.2 < 0.0001 [0.0000; 0.0617]; tau.2 < 0.0001 [0.0000; 0.2484] (within cluster)
##  I^2 = 31.2% [0.0%; 60.0%]; H = 1.21 [1.00; 1.58]
##
## Quantifying residual heterogeneity (with 95%-CIs):
##  tau^2 = 0; tau = 0; I^2 = 0.0%; H = 1.00
##
## Test of heterogeneity:
##      Q d.f. p-value
##  27.61   19  0.0911
##
## Results for subgroups (random effects model):
##                              k     COR            95%-CI   tau^2     tau    Q
## inhibitory.measure = SICI   15 -0.0949 [-0.1925; 0.0045] <0.0001 <0.0001 8.22
## inhibitory.measure = cSP     2  0.1428 [-0.1608; 0.4217] <0.0001 <0.0001 0.02
## inhibitory.measure = iSP     2  0.4652 [ 0.2172; 0.6568] <0.0001 <0.0001 0.03
## inhibitory.measure = LICI    1  0.0490 [-0.2786; 0.3664] <0.0001 <0.0001 0.00
##                             I^2
## inhibitory.measure = SICI 0.0%
## inhibitory.measure = cSP  0.0%
## inhibitory.measure = iSP  0.0%
## inhibitory.measure = LICI   --
##
## Test for subgroup differences (random effects model):
##                    Q d.f. p-value
## Between groups 6.45    3  0.0045
##
## Details of meta-analysis methods:
## - Inverse variance method (three-level model)
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Profile-Likelihood method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Random effects confidence interval based on t-distribution (df = 19)
## - Fisher's z transformation of correlations
```

ADD TEXT

```
forest(inhibition_measure_type,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)
```



Inhibition measure type: Save plot

```
library(svglite)

svglite("forest_plot_inhibition-measure.svg", height = 10, width = 14)

forest(inhibition_measure_type,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)

dev.off()
```

```
## pdf
##   2
```

Facilitation Measure

```r
facilitation_r$facilitatory.measure <- as.factor(facilitation_r$facilitatory.measure)

facilitation_type <- rma.mv(yi = z,
                      V = var.z,
                      slab = author,
                      data = facilitation_r,
                      random = ~ 1 | author/cor_id,
                      test = "t",
                      method = "REML",
                      mods = ~ facilitatory.measure)

summary(facilitation_type)
```

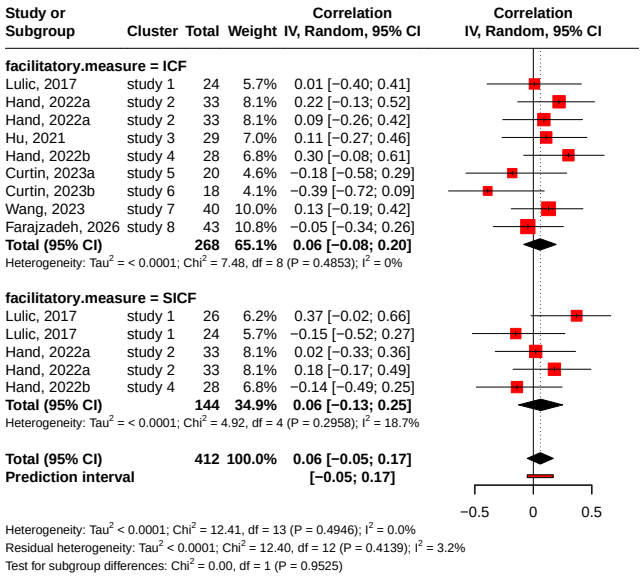
```
##
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##   2.1854   -4.3708    3.6292    5.5688    9.3435
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0000  0.0000      8     no         author
## sigma^2.2  0.0000  0.0000     14     no  author/cor_id
##
## Test for Residual Heterogeneity:
## QE(df = 12) = 12.4030, p-val = 0.4139
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 12) = 0.0037, p-val = 0.9525
##
## Model Results:
##
##                           estimate      se    tval  df    pval    ci.lb   ci.ub
## intrcpt                     0.0577  0.0644  0.8954  12  0.3882  -0.0827  0.1980
## facilitatory.measureSICF    0.0066  0.1091  0.0608  12  0.9525  -0.2311  0.2443
##
## intrcpt
## facilitatory.measureSICF
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
facilitation_measure_type <- update(m.facilitation, subgroup = facilitatory.measure, tau.common = TRUE)
facilitation_measure_type
```

```
## Review:     Sensitivity meta-analysis using metacor
##
## Number of studies: n = 8
## Number of estimates: k = 14
## Number of observations: o = 412
##
```

```
##                          COR            95%-CI    t p-value
## Random effects model 0.0599 [-0.0523; 0.1706] 1.15  0.2693
##
## Quantifying heterogeneity (with 95%-CIs):
##  tau^2.1 = 0      [0.0000; 0.0410]; tau.1 = 0      [0.0000; 0.2025] (between cluster)
##  tau^2.2 < 0.0001 [0.0000; 0.0439]; tau.2 < 0.0001 [0.0000; 0.2095] (within cluster)
##  I^2 = 0.0% [0.0%; 55.0%]; H = 1.00 [1.00; 1.49]
##
## Quantifying residual heterogeneity (with 95%-CIs):
##  tau^2 < 0.0001; tau = 0.0003; I^2 = 3.2%; H = 1.02
##
## Test of heterogeneity:
##      Q d.f. p-value
##  12.41   13  0.4946
##
## Results for subgroups (random effects model):
##                               k    COR            95%-CI   tau^2     tau    Q
## facilitatory.measure = ICF    9 0.0576 [-0.0825; 0.1955] <0.0001 <0.0001 7.48
## facilitatory.measure = SICF   5 0.0642 [-0.1268; 0.2507] <0.0001 <0.0001 4.92
##                               I^2
## facilitatory.measure = ICF   0.0%
## facilitatory.measure = SICF 18.7%
##
## Test for subgroup differences (random effects model):
##                    Q d.f. p-value
## Between groups 0.00    1  0.9525
##
## Details of meta-analysis methods:
## - Inverse variance method (three-level model)
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Profile-Likelihood method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Random effects confidence interval based on t-distribution (df = 13)
## - Fisher's z transformation of correlations
```

```r
forest(facilitation_measure_type,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)
```



Facilitation by tms measure type: Save plot

```
library(svglite)

svglite("forest_plot_facilitation-type.svg", height = 7, width = 14)

forest(facilitation_measure_type,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)

dev.off()
```

```
## pdf
##   2
```

Physical Activity Objectivity for Excitability

```
excitability_r$PA_objectivity <- as.factor(excitability_r$PA_objectivity)

excitability_PA_type <- rma.mv(yi = z,
                      V = var.z,
                      slab = author,
                      data = excitability_r,
                      random = ~ 1 | author/cor_id,
                      test = "t",
                      method = "REML",
```

```
                     mods = ~ PA_objectivity)

summary(excitability_PA_type)
```

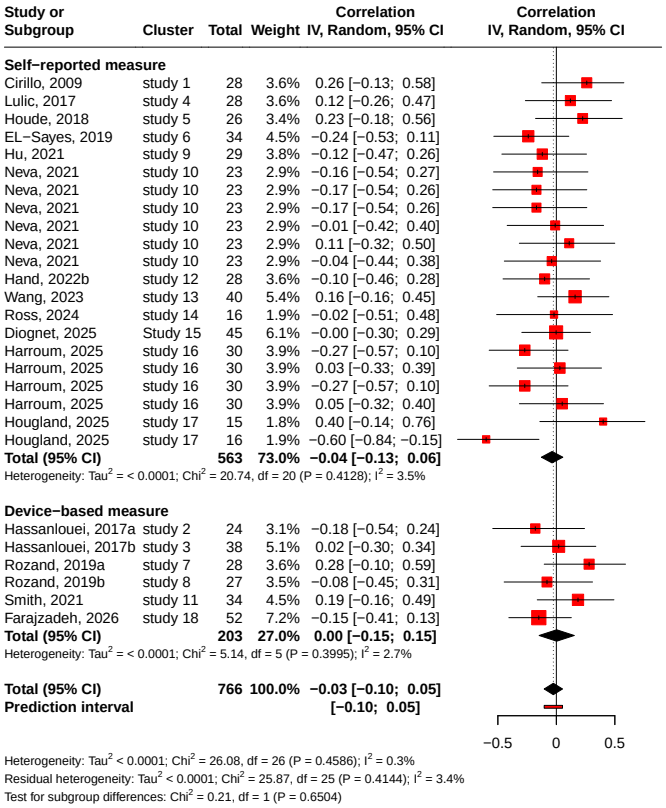
```
##
## Multivariate Meta-Analysis Model (k = 27; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##   3.7551   -7.5101    0.4899    5.3654    2.4899
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0000  0.0000     18     no         author
## sigma^2.2  0.0000  0.0000     27     no  author/cor_id
##
## Test for Residual Heterogeneity:
## QE(df = 25) = 25.8722, p-val = 0.4144
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 25) = 0.2105, p-val = 0.6504
##
## Model Results:
##
##                                     estimate      se     tval  df    pval
## intrcpt                               0.0029  0.0735   0.0399  25  0.9685
## PA_objectivitySelf-reported measure  -0.0395  0.0861  -0.4588  25  0.6504
##                                       ci.lb   ci.ub
## intrcpt                             -0.1485  0.1544
## PA_objectivitySelf-reported measure -0.2167  0.1378
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
excitability_PA_type <- update(m.excitability, subgroup = PA_objectivity, tau.common = TRUE)
excitability_PA_type
```

```
## Review:     Sensitivity meta-analysis using metacor
##
## Number of studies: n = 18
## Number of estimates: k = 27
## Number of observations: o = 766
##
##                          COR            95%-CI     t p-value
## Random effects model -0.0259 [-0.1040; 0.0526] -0.68  0.5041
##
## Quantifying heterogeneity (with 95%-CIs):
##  tau^2.1 < 0.0001 [0.0000; 0.0236]; tau.1 < 0.0001 [0.0000; 0.1537] (between cluster)
##  tau^2.2 < 0.0001 [0.0000; 0.0280]; tau.2 < 0.0001 [0.0000; 0.1675] (within cluster)
##  I^2 = 0.3% [0.0%; 42.7%]; H = 1.00 [1.00; 1.32]
##
## Quantifying residual heterogeneity (with 95%-CIs):
```

```
##  tau^2 < 0.0001; tau = 0.0008; I^2 = 3.4%; H = 1.02
##
## Test of heterogeneity:
##      Q d.f. p-value
##  26.08   26  0.4586
##
## Results for subgroups (random effects model):
##                                          k     COR            95%-CI   tau^2
## PA_objectivity = Self-reported measure  21 -0.0365 [-0.1279; 0.0555] <0.0001
## PA_objectivity = Device-based measure    6  0.0029 [-0.1474; 0.1531] <0.0001
##                                            tau     Q  I^2
## PA_objectivity = Self-reported measure <0.0001 20.74 3.5%
## PA_objectivity = Device-based measure  <0.0001  5.14 2.7%
##
## Test for subgroup differences (random effects model):
##                    Q d.f. p-value
## Between groups 0.21    1  0.6504
##
## Details of meta-analysis methods:
## - Inverse variance method (three-level model)
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Profile-Likelihood method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Random effects confidence interval based on t-distribution (df = 26)
## - Fisher's z transformation of correlations
```

```r
forest(excitability_PA_type,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       print.subgroup.name = FALSE,
       addrows.below.overall = 2)
```



excitability by PA measure type: Save plot

```
library(svglite)

svglite("forest_plot_excitability-PA-type.svg", height = 7, width = 14)

forest(excitability_PA_type,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)

dev.off()
```

```
## pdf
##   2
```

Physical Activity Objectivity for Inhibition

```r
inhibition_r$PA_objectivity <- as.factor(inhibition_r$PA_objectivity)

inhibition_PA_type <- rma.mv(yi = z,
                      V = var.z,
                      slab = author,
                      data = inhibition_r,
                      random = ~ 1 | author/cor_id,
                      test = "t",
                      method = "REML",
                      mods = ~ PA_objectivity)

summary(inhibition_PA_type)
```

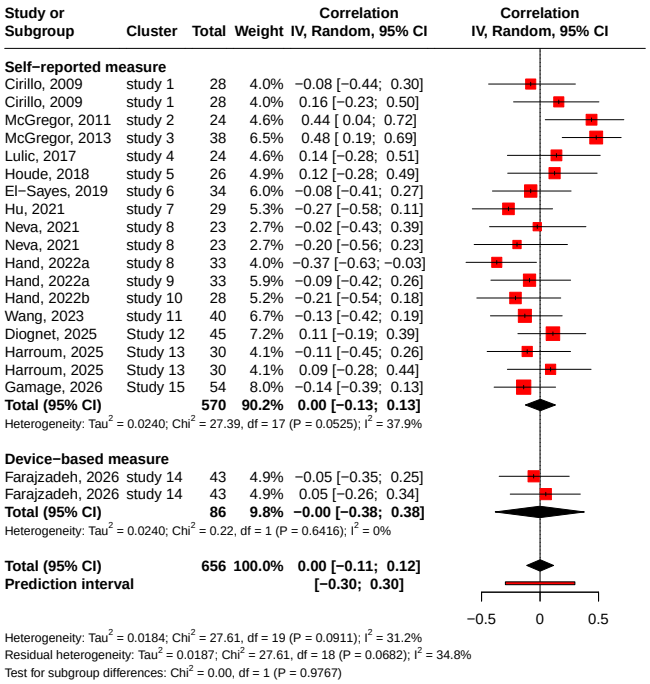
```
##
## Multivariate Meta-Analysis Model (k = 20; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##   1.1298   -2.2596    5.7404    9.3018    8.8173
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0239  0.1545     15     no         author
## sigma^2.2  0.0000  0.0000     20     no  author/cor_id
##
## Test for Residual Heterogeneity:
## QE(df = 18) = 27.6100, p-val = 0.0682
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 18) = 0.0003, p-val = 0.9853
##
## Model Results:
##
##                                      estimate      se     tval  df    pval
## intrcpt                               -0.0030  0.1907  -0.0158  18  0.9876
## PA_objectivitySelf-reported measure    0.0037  0.2003   0.0187  18  0.9853
##                                        ci.lb   ci.ub
## intrcpt                               -0.4037  0.3977
## PA_objectivitySelf-reported measure  -0.4171  0.4246
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
inhibition_PA_type <- update(m.inhibition, subgroup = PA_objectivity, tau.common = TRUE)
inhibition_PA_type
```

```
## Review:     Sensitivity meta-analysis using metacor
##
## Number of studies: n = 15
## Number of estimates: k = 20
## Number of observations: o = 656
##
```

```
##                          COR            95%-CI    t p-value
## Random effects model 0.0008 [-0.1140; 0.1155] 0.01  0.9888
##
## Quantifying heterogeneity (with 95%-CIs):
##  tau^2.1 = 0.0184 [0.0000; 0.0842]; tau.1 = 0.1356 [0.0000; 0.2901] (between cluster)
##  tau^2.2 < 0.0001 [0.0000; 0.0617]; tau.2 < 0.0001 [0.0000; 0.2484] (within cluster)
##  I^2 = 31.2% [0.0%; 60.0%]; H = 1.21 [1.00; 1.58]
##
## Quantifying residual heterogeneity (with 95%-CIs):
##  tau^2 = 0.0187; tau = 0.1367; I^2 = 34.8%; H = 1.24
##
## Test of heterogeneity:
##      Q d.f. p-value
##  27.61   19  0.0911
##
## Results for subgroups (random effects model):
##                                          k     COR            95%-CI  tau^2
## PA_objectivity = Self-reported measure  18  0.0029 [-0.1256; 0.1314] 0.0240
## PA_objectivity = Device-based measure    2 -0.0030 [-0.3837; 0.3786] 0.0240
##                                           tau     Q   I^2
## PA_objectivity = Self-reported measure 0.1549 27.39 37.9%
## PA_objectivity = Device-based measure  0.1549  0.22  0.0%
##
## Test for subgroup differences (random effects model):
##                    Q d.f. p-value
## Between groups 0.00    1  0.9767
##
## Details of meta-analysis methods:
## - Inverse variance method (three-level model)
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Profile-Likelihood method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Random effects confidence interval based on t-distribution (df = 19)
## - Fisher's z transformation of correlations
```

```
forest(inhibition_PA_type,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       print.subgroup.name = FALSE,
       addrows.below.overall = 2)
```



Inhibition by PA measure type: Save plot

```
library(svglite)

svglite("forest_plot_inhibition-PA-type.svg", height = 7, width = 14)

forest(inhibition_PA_type,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)

dev.off()
```

```
## pdf
##   2
```

Physical Activity Objectivity for Facilitation

```
facilitation_r$PA_objectivity <- as.factor(facilitation_r$PA_objectivity)

facilitation_PA_type <- rma.mv(yi = z,
                     V = var.z,
                     slab = author,
                     data = facilitation_r,
```

```
                      random = ~ 1 | author/cor_id,
                      test = "t",
                      method = "REML",
                      mods = ~ PA_objectivity)

summary(facilitation_PA_type)
```

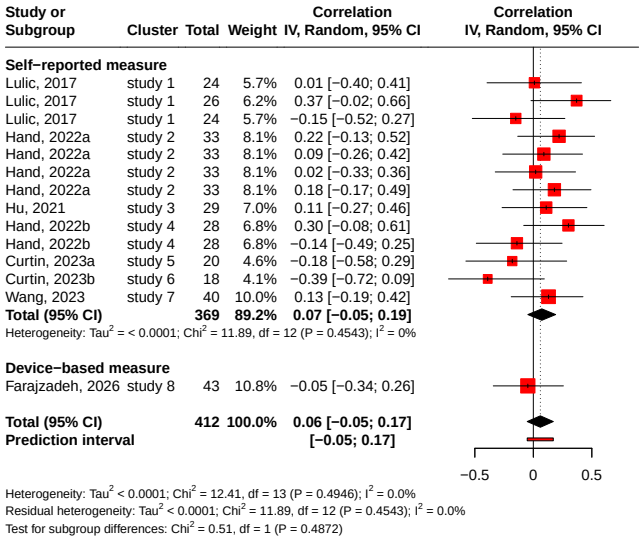
```
##
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##   2.2479   -4.4957    3.5043    5.4439    9.2186
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0000  0.0000      8     no         author
## sigma^2.2  0.0000  0.0000     14     no  author/cor_id
##
## Test for Residual Heterogeneity:
## QE(df = 12) = 11.8930, p-val = 0.4543
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 12) = 0.5137, p-val = 0.4872
##
## Model Results:
##
##                                     estimate      se     tval  df    pval
## intrcpt                              -0.0470  0.1581  -0.2975  12  0.7712
## PA_objectivitySelf-reported measure   0.1200  0.1674   0.7167  12  0.4872
##                                        ci.lb   ci.ub
## intrcpt                              -0.3915  0.2975
## PA_objectivitySelf-reported measure  -0.2448  0.4848
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
facilitation_PA_type <- update(m.facilitation, subgroup = PA_objectivity, tau.common = TRUE)
facilitation_PA_type
```

```
## Review:     Sensitivity meta-analysis using metacor
##
## Number of studies: n = 8
## Number of estimates: k = 14
## Number of observations: o = 412
##
##                          COR            95%-CI    t p-value
## Random effects model 0.0599 [-0.0523; 0.1706] 1.15  0.2693
##
## Quantifying heterogeneity (with 95%-CIs):
##  tau^2.1 = 0      [0.0000; 0.0410]; tau.1 = 0      [0.0000; 0.2025] (between cluster)
##  tau^2.2 < 0.0001 [0.0000; 0.0439]; tau.2 < 0.0001 [0.0000; 0.2095] (within cluster)
```

```
##  I^2 = 0.0% [0.0%; 55.0%]; H = 1.00 [1.00; 1.49]
##
## Quantifying residual heterogeneity (with 95%-CIs):
##  tau^2 < 0.0001; tau = 0.0022; I^2 = 0.0%; H = 1.00
##
## Test of heterogeneity:
##      Q d.f. p-value
##  12.41   13  0.4946
##
## Results for subgroups (random effects model):
##                                          k     COR            95%-CI   tau^2
## PA_objectivity = Self-reported measure  13  0.0728 [-0.0469; 0.1905] <0.0001
## PA_objectivity = Device-based measure    1 -0.0470 [-0.3727; 0.2890] <0.0001
##                                            tau     Q  I^2
## PA_objectivity = Self-reported measure <0.0001 11.89 0.0%
## PA_objectivity = Device-based measure  <0.0001  0.00   --
##
## Test for subgroup differences (random effects model):
##                   Q d.f. p-value
## Between groups 0.51    1  0.4872
##
## Details of meta-analysis methods:
## - Inverse variance method (three-level model)
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Profile-Likelihood method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Random effects confidence interval based on t-distribution (df = 13)
## - Fisher's z transformation of correlations
```

```
forest(facilitation_PA_type,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       print.subgroup.name = FALSE,
       addrows.below.overall = 2)
```



Facilitation by tms measure type: Save plot

```
library(svglite)

svglite("forest_plot_facilitation-type.svg", height = 7, width = 14)

forest(facilitation_PA_type,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)

dev.off()
```

```
## pdf
##   2
```

Meta Regressions

Excitability

Meta-regressions by proportion of women

```
m.cor.reg.excitability.women <- metareg(m.excitability, ~Prop_women)
```

```
## Warning: 1 row with NAs omitted from model fitting.
```

```
m.cor.reg.excitability.women
```

```
##
## Multivariate Meta-Analysis Model (k = 26; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0000  0.0000     17     no       .id
## sigma^2.2  0.0000  0.0000     26     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 24) = 25.0431, p-val = 0.4034
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 24) = 1.0140, p-val = 0.3240
##
## Model Results:
##
##             estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt       0.1232  0.1548   0.7964  24  0.4336  -0.1962  0.4426
## Prop_women   -0.2870  0.2850  -1.0070  24  0.3240  -0.8753  0.3013
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Meta-regressions by age

```
m.cor.reg.excitability.age <- metareg(m.excitability, ~Age)
m.cor.reg.excitability.age
```

```
##
## Multivariate Meta-Analysis Model (k = 27; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0000  0.0000     18     no       .id
## sigma^2.2  0.0000  0.0000     27     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 25) = 24.8219, p-val = 0.4724
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 25) = 1.2608, p-val = 0.2722
##
## Model Results:
##
##          estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt   -0.1111  0.0850  -1.3076  25  0.2029  -0.2861  0.0639
## Age        0.0026  0.0023   1.1229  25  0.2722  -0.0022  0.0073
```

```
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Meta-regressions by study quality

```
m.cor.reg.excitability.quality <- metareg(m.excitability, ~Q_score)
m.cor.reg.excitability.quality
```

```
##
## Multivariate Meta-Analysis Model (k = 27; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0000  0.0000     18     no        .id
## sigma^2.2  0.0000  0.0000     27     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 25) = 25.5360, p-val = 0.4327
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 25) = 0.5467, p-val = 0.4666
##
## Model Results:
##
##          estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt    0.1966  0.3033   0.6481  25  0.5228  -0.4281  0.8214
## Q_score   -0.0155  0.0209  -0.7394  25  0.4666  -0.0586  0.0276
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Inhibition

Meta-regressions by proportion of women

```
m.cor.reg.inhibition.women <- metareg(m.inhibition, ~Prop_women)
```

```
## Warning: 1 row with NAs omitted from model fitting.
```

```
m.cor.reg.inhibition.women
```

```
##
## Multivariate Meta-Analysis Model (k = 19; method: REML)
##
## Variance Components:
##
```

```
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0216  0.1471     14     no       .id
## sigma^2.2  0.0000  0.0000     19     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 17) = 25.5489, p-val = 0.0831
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 17) = 1.0090, p-val = 0.3292
##
## Model Results:
##
##             estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt      -0.0259  0.0623  -0.4161  17  0.6825  -0.1574  0.1055
## Prop_women    0.0038  0.0038   1.0045  17  0.3292  -0.0042  0.0118
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Meta-regressions by age

```
m.cor.reg.inhibition.age <- metareg(m.inhibition, ~Age)
m.cor.reg.inhibition.age
```

```
##
## Multivariate Meta-Analysis Model (k = 20; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0529  0.2300     15     no       .id
## sigma^2.2  0.0000  0.0000     20     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 7) = 9.3363, p-val = 0.2294
##
## Test of Moderators (coefficients 2:13):
## F(df1 = 12, df2 = 7) = 0.6036, p-val = 0.7890
##
## Model Results:
##
##           estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt    -0.2769  0.3023  -0.9159   7  0.3902  -0.9916  0.4379
## Age21       0.2981  0.3708   0.8040   7  0.4478  -0.5786  1.1748
## Age23       0.0637  0.4293   0.1484   7  0.8862  -0.9514  1.0788
## Age23.1     0.3873  0.4100   0.9447   7  0.3763  -0.5821  1.3568
## Age24.4     0.3175  0.4053   0.7833   7  0.4591  -0.6409  1.2759
## Age25       0.1461  0.4139   0.3531   7  0.7344  -0.8326  1.1248
## Age27       0.2565  0.3912   0.6557   7  0.5330  -0.6686  1.1816
## Age27(6)    0.2668  0.4035   0.6612   7  0.5297  -0.6873  1.2208
## Age32       0.0375  0.3667   0.1023   7  0.9214  -0.8296  0.9047
## Age32.6     0.2739  0.3960   0.6916   7  0.5114  -0.6624  1.2101
```

```
## Age52.1     0.7998  0.4158   1.9239   7  0.0958  -0.1832  1.7829  .
## Age69       0.3929  0.3654   1.0753   7  0.3179  -0.4711  1.2569
## Age70       0.4015  0.4333   0.9266   7  0.3850  -0.6231  1.4261
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Meta-regressions by study quality

```
m.cor.reg.inhibition.quality <- metareg(m.inhibition, ~Q_score)
m.cor.reg.inhibition.quality
```

```
##
## Multivariate Meta-Analysis Model (k = 20; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0171  0.1307     15     no       .id
## sigma^2.2  0.0000  0.0000     20     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 18) = 24.5054, p-val = 0.1392
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 18) = 2.6653, p-val = 0.1199
##
## Model Results:
##
##          estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt   -0.5894  0.3653  -1.6135  18  0.1240  -1.3569  0.1780
## Q_score    0.0430  0.0263   1.6326  18  0.1199  -0.0123  0.0982
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Facilitation

Meta-regressions by proportion of women

```
m.cor.reg.facilitation.women <- metareg(m.facilitation, ~Prop_women)
m.cor.reg.facilitation.women
```

```
##
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
```

```
## sigma^2.1  0.0000  0.0000      8     no       .id
## sigma^2.2  0.0000  0.0000     14     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 12) = 11.6963, p-val = 0.4704
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 12) = 0.7104, p-val = 0.4158
##
## Model Results:
##
##             estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt       0.1449  0.1134   1.2781  12  0.2254  -0.1021  0.3920
## Prop_women   -0.1909  0.2265  -0.8429  12  0.4158  -0.6845  0.3026
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Meta-regressions by age

```
m.cor.reg.facilitation.age <- metareg(m.facilitation, ~Age)
m.cor.reg.facilitation.age
```

```
##
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0000  0.0000      8     no       .id
## sigma^2.2  0.0000  0.0000     14     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 12) = 9.8929, p-val = 0.6254
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 12) = 2.5138, p-val = 0.1388
##
## Model Results:
##
##          estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt    0.3230  0.1738   1.8580  12  0.0879  -0.0558  0.7018  .
## Age       -0.0093  0.0059  -1.5855  12  0.1388  -0.0221  0.0035
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Meta-regressions by study quality

```r
m.cor.reg.facilitation.quality <- metareg(m.facilitation, ~Q_score)
m.cor.reg.facilitation.quality
```

```
##
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0000  0.0000      8     no        .id
## sigma^2.2  0.0000  0.0000     14     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 12) = 9.8681, p-val = 0.6275
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 12) = 2.5386, p-val = 0.1371
##
## Model Results:
##
##          estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt    0.7237  0.4198   1.7239  12  0.1104  -0.1910  1.6384
## Q_score   -0.0519  0.0326  -1.5933  12  0.1371  -0.1229  0.0191
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

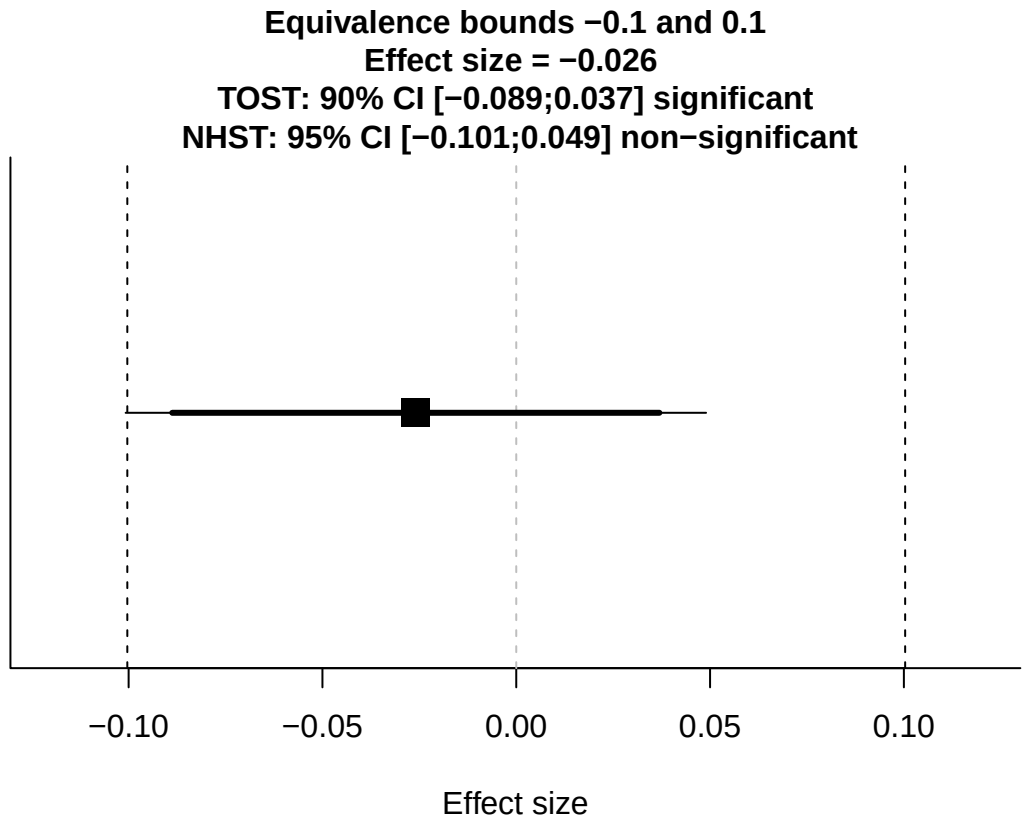
Equivalence Testing

Excitability

```r
# Define SESOI in r
sesoi_r <- 0.1

# Convert r to Fisher's z
low_eq_z  <- atanh(-sesoi_r)
high_eq_z <- atanh( sesoi_r)

TOSTmeta(
  ES = -0.0259,
  se = 0.0382,
  low_eqbound  = low_eq_z,
  high_eqbound =  high_eq_z,
  alpha = 0.05
)
```




```
## TOST results:
## Z-value lower bound: 1.95    p-value lower bound: 0.026
## Z-value upper bound: -3.30   p-value upper bound: 0.0005
##
## Equivalence bounds (Cohen's d):
## low eqbound: -0.1003
## high eqbound: 0.1003
##
## TOST confidence interval:
## lower bound 90% CI: -0.089
## upper bound 90% CI:  0.037
##
## NHST confidence interval:
## lower bound 95% CI: -0.101
## upper bound 95% CI:  0.049
##
## Equivalence Test Result:
## The equivalence test was significant, Z = 1.949, p = 0.0257, given equivalence bounds of -0.100 and (

##

##
## Null Hypothesis Test Result:
## The null hypothesis test was non-significant, Z = -0.678, p = 0.498, given an alpha of 0.05.

##
```

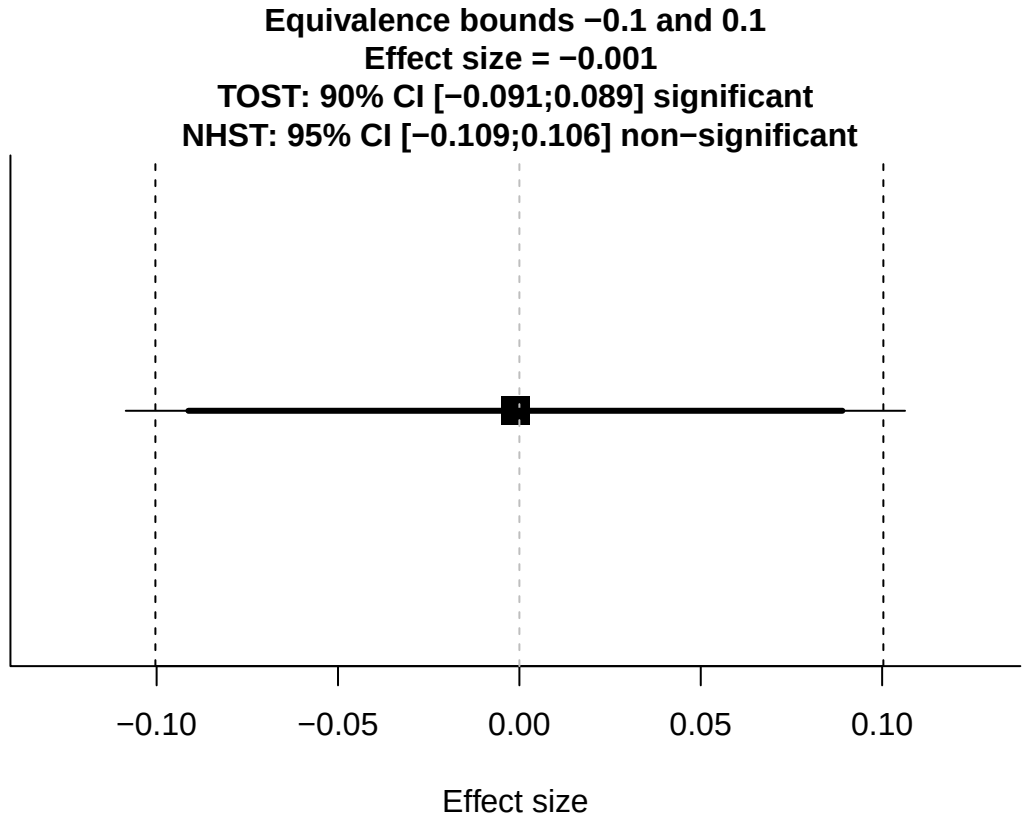
```
## NHST: don't reject null significance hypothesis that the effect is equal to 0
## TOST: reject null equivalence hypothesis
```

Inhibition

```r
# Define SESOI in r
sesoi_r <- 0.1

# Convert r to Fisher's z
low_eq_z  <- atanh(-sesoi_r)
high_eq_z <- atanh( sesoi_r)

TOSTmeta(
  ES = -0.0011,
  se = 0.0548,
  low_eqbound  = low_eq_z,
  high_eqbound =  high_eq_z,
  alpha = 0.05
)
```




```
## TOST results:
## Z-value lower bound: 1.81    p-value lower bound: 0.035
## Z-value upper bound: -1.85   p-value upper bound: 0.032
##
```

```
## Equivalence bounds (Cohen's d):
## low eqbound: -0.1003
## high eqbound: 0.1003
##
## TOST confidence interval:
## lower bound 90% CI: -0.091
## upper bound 90% CI:  0.089
##
## NHST confidence interval:
## lower bound 95% CI: -0.109
## upper bound 95% CI:  0.106
##
## Equivalence Test Result:
## The equivalence test was significant, Z = 1.811, p = 0.0351, given equivalence bounds of -0.100 and (

##


##
## Null Hypothesis Test Result:
## The null hypothesis test was non-significant, Z = -0.0201, p = 0.984, given an alpha of 0.05.

##

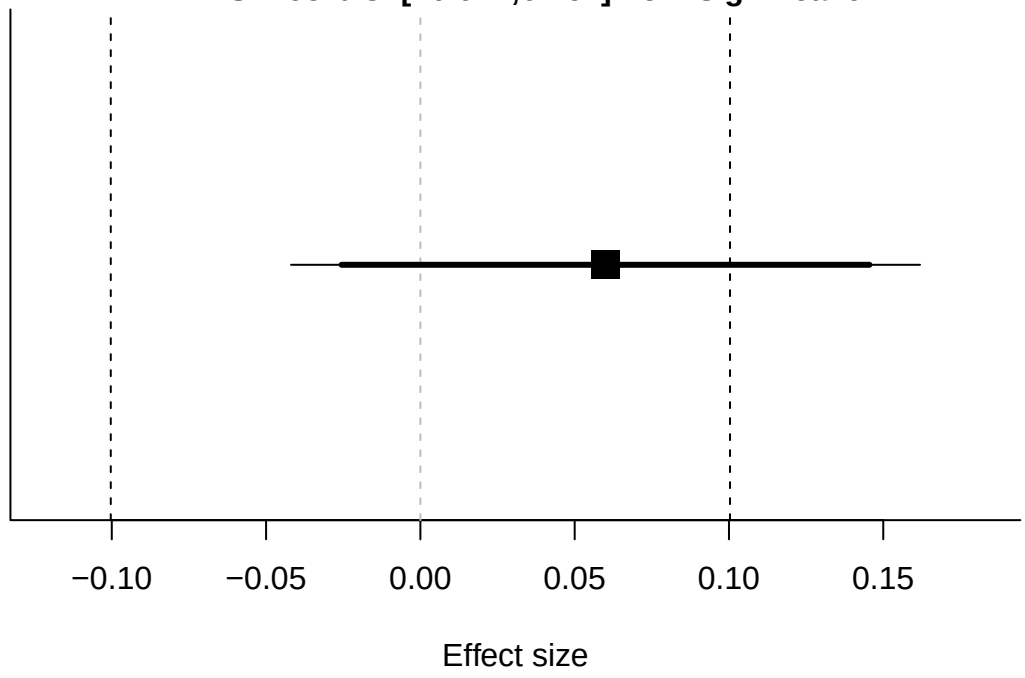
## NHST: don't reject null significance hypothesis that the effect is equal to 0
## TOST: reject null equivalence hypothesis
```

Facilitation

```r
# Define SESOI in r
sesoi_r <- 0.1

# Convert r to Fisher's z
low_eq_z  <- atanh(-sesoi_r)
high_eq_z <- atanh( sesoi_r)

TOSTmeta(
  ES = 0.0600,
  se = 0.0520,
  low_eqbound  = low_eq_z,
  high_eqbound =  high_eq_z,
  alpha = 0.05
)
```




```
## TOST results:
## Z-value lower bound: 3.08    p-value lower bound: 0.001
## Z-value upper bound: -0.776  p-value upper bound: 0.219
##
## Equivalence bounds (Cohen's d):
## low eqbound: -0.1003
## high eqbound: 0.1003
##
## TOST confidence interval:
## lower bound 90% CI: -0.026
## upper bound 90% CI:  0.146
##
## NHST confidence interval:
## lower bound 95% CI: -0.042
## upper bound 95% CI:  0.162
##
## Equivalence Test Result:
## The equivalence test was non-significant, Z = -0.776, p = 0.219, given equivalence bounds of -0.100 a

##

##
## Null Hypothesis Test Result:
## The null hypothesis test was non-significant, Z = 1.154, p = 0.249, given an alpha of 0.05.

##
```

```
## NHST: don't reject null significance hypothesis that the effect is equal to 0
## TOST: don't reject null equivalence hypothesis
```

Figure

```
library(ggplot2)

# Define SESOI
sesoi_r <- 0.1

# Equivalence bounds in Fisher's z
low_eq_z  <- atanh(-sesoi_r)
high_eq_z <- atanh(sesoi_r)

# Results from the three meta-analyses
results <- data.frame(
  Measure = c("Corticospinal Excitability", "Inhibitory Processes", "Facilitatory Processes"),
  ES = c(-0.0259, -0.0011, 0.0600),
  SE = c(0.0382, 0.0548, 0.0520)
)

results$Measure <- factor(
  results$Measure,
  levels = c("Facilitatory Processes", "Inhibitory Processes", "Corticospinal Excitability")
)

results$y <- c(1.04, 1.02, 1)

# Critical values
z90 <- qnorm(0.95)   # 90% CI

# Calculate CIs on Fisher's z scale and transform back to r

# 90% CI for equivalence testing
results$lower90 <- tanh(
  atanh(results$ES) - z90 * results$SE
)

results$upper90 <- tanh(
  atanh(results$ES) + z90 * results$SE
)

# Plot
ggplot(results, aes(x = ES, y = y)) +

  # 90% CI - equivalence test (thick)
  geom_errorbarh(
    aes(xmin = lower90, xmax = upper90),
    height = 0,
    linewidth = 1.6
  ) +
```

```
  # Point estimate
  geom_point(size = 3) +

  # Equivalence bounds
  geom_vline(
    xintercept = c(-sesoi_r, sesoi_r),
    linetype = "dashed"
  ) +

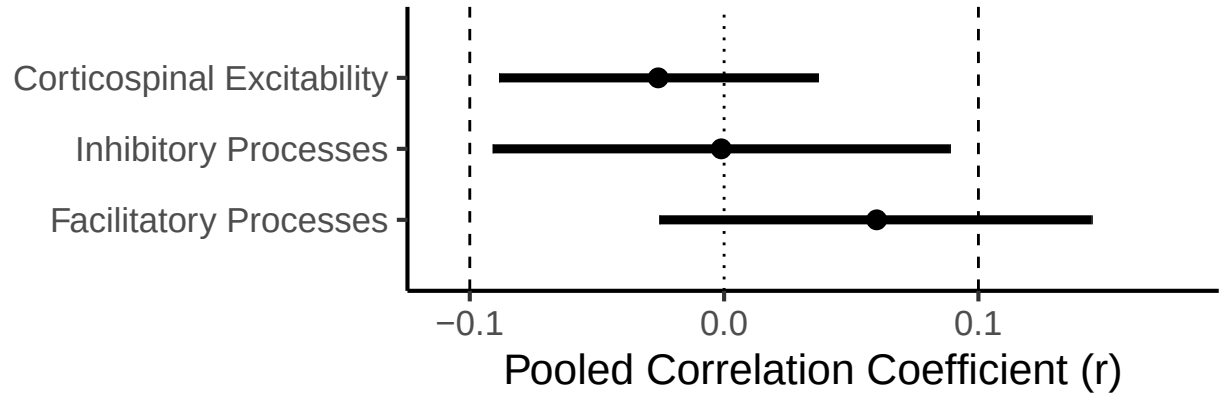
  # Null value
  geom_vline(
    xintercept = 0,
    linetype = "dotted"
  ) +

scale_y_continuous(
  breaks = results$y,
  labels = results$Measure,
  limits = c(0.98, 1.06),
  expand = c(0, 0)
) +

  labs(
    x = "Pooled Correlation Coefficient (r)",
    y = NULL
  ) +

  theme_classic(base_size = 16) +
  theme(
  aspect.ratio = 0.35
) +

  coord_cartesian(
    xlim = c(-0.11, 0.18)
  )
```



Publication bias analysis

Small-study effects

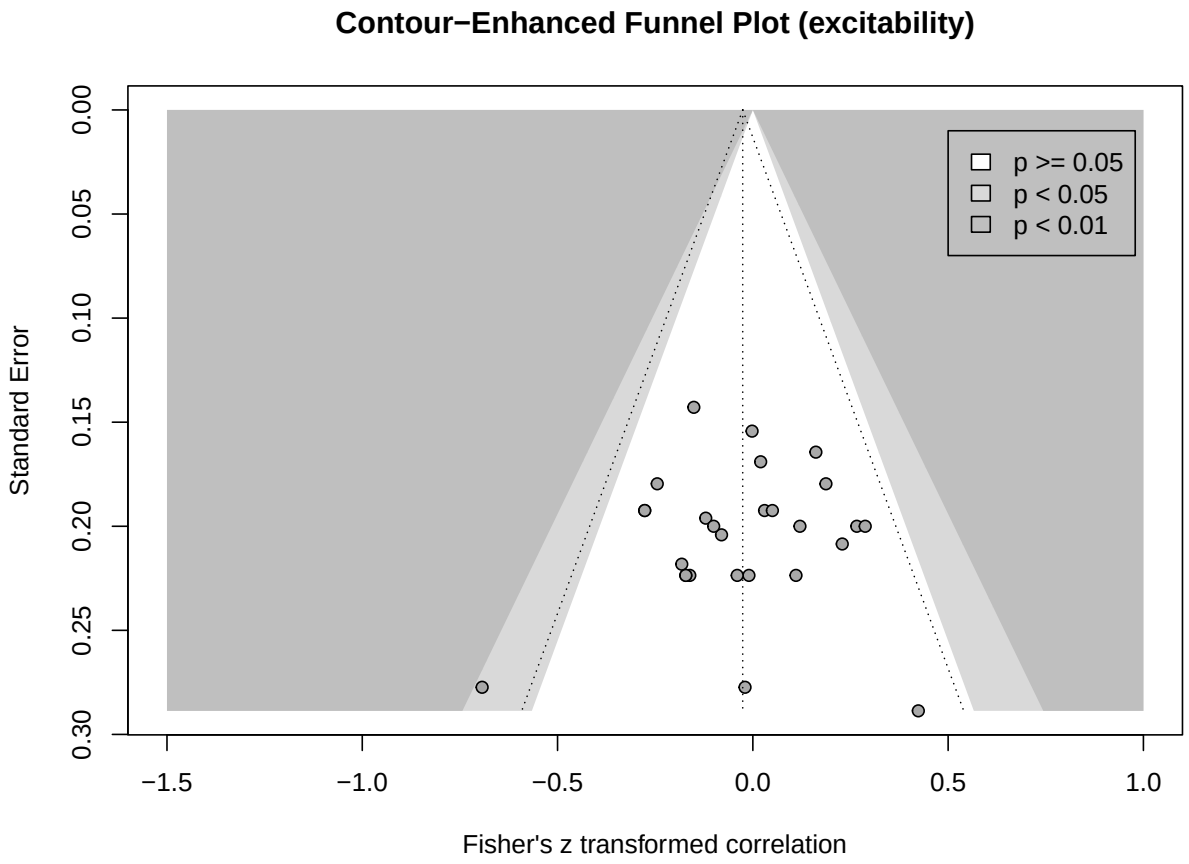
Funnel plot

```
# Define fill colors for contour
col.contour = c("gray85", "gray75")

# Funnel plot
meta::funnel(m.excitability,
             xlim = c(-1.5, 1),
             contour = c(0.95, 0.99),
             col.contour = col.contour)

# legend
legend(x =0.5, y = 0.01,
       legend = c("p  0.05", "p < 0.05", "p < 0.01"),
      fill = c("white", col.contour))

# title
title("Contour-Enhanced Funnel Plot (excitability)")
```

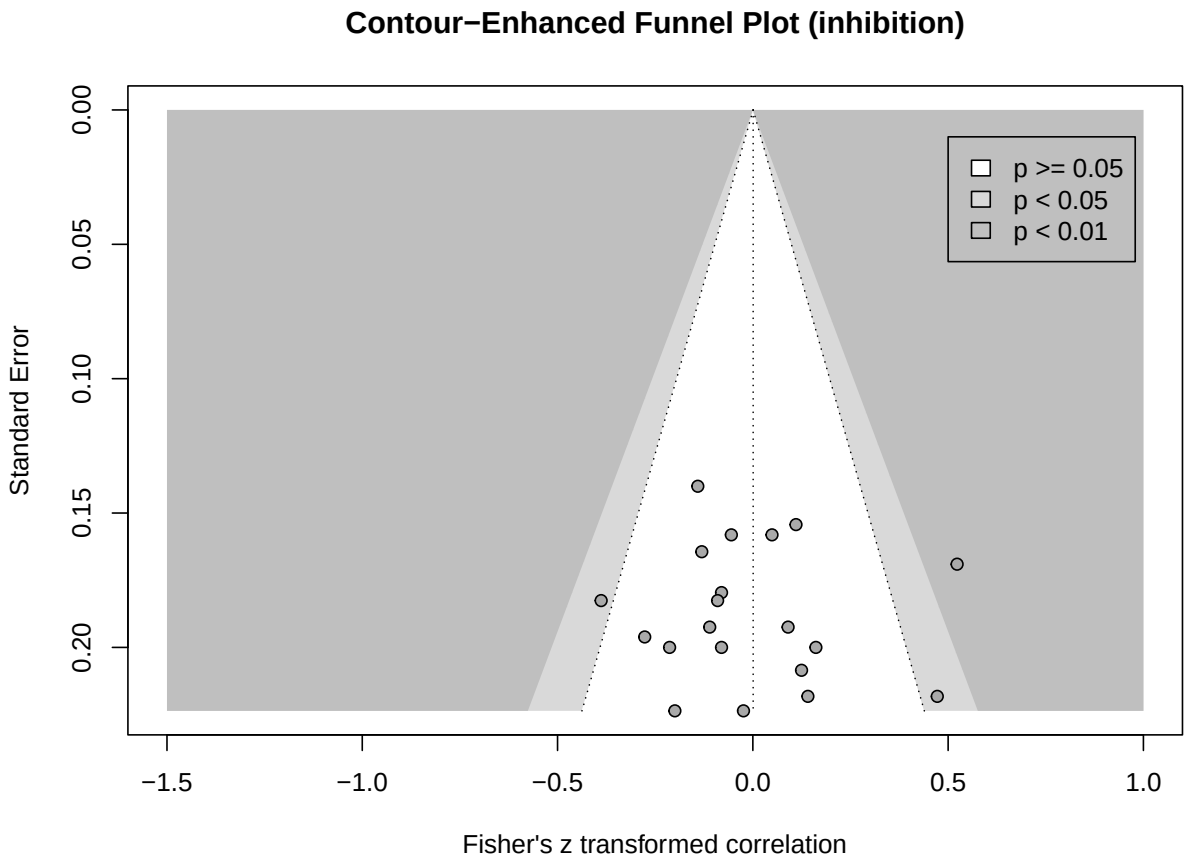



```r
# Define fill colors for contour
col.contour = c("gray85", "gray75")

# Funnel plot
meta::funnel(m.inhibition,
             xlim = c(-1.5, 1),
             contour = c(0.95, 0.99),
             col.contour = col.contour)

# legend
legend(x =0.5, y = 0.01,
       legend = c("p  0.05", "p < 0.05", "p < 0.01"),
      fill = c("white", col.contour))

# title
title("Contour-Enhanced Funnel Plot (inhibition)")
```




```
png(file = "Funnel Plot.png", width = 2500, height = 2000, res = 300)

# Define fill colors for contour
col.contour = c("gray85", "gray75")

# Funnel plot
meta::funnel(m.inhibition,
             xlim = c(-1.5, 1),
             contour = c(0.95, 0.99),
             col.contour = col.contour)

# legend
legend(x =0.5, y = 0.01,
       legend = c("p   0.05", "p < 0.05", "p < 0.01"),
      fill = c("white", col.contour))

# title
title("Contour-Enhanced Funnel Plot (tms and Physical Activity)")

dev.off()
```

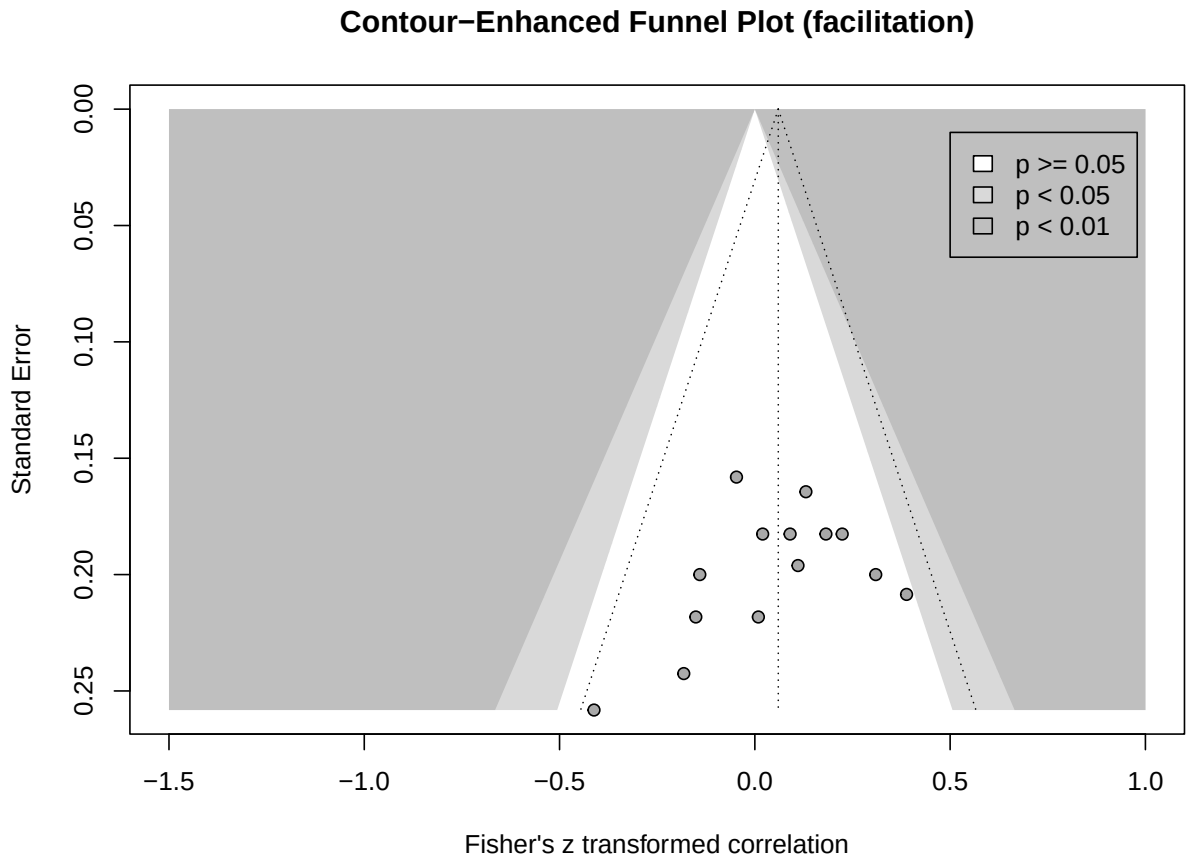
```
## pdf
##   2
```

```r
# Define fill colors for contour
col.contour = c("gray85", "gray75")

# Funnel plot
meta::funnel(m.facilitation,
             xlim = c(-1.5, 1),
             contour = c(0.95, 0.99),
             col.contour = col.contour)

# legend
legend(x =0.5, y = 0.01,
       legend = c("p  0.05", "p < 0.05", "p < 0.01"),
      fill = c("white", col.contour))

# title
title("Contour-Enhanced Funnel Plot (facilitation)")
```



Egger's test

ADD TEXT

```r
test.egger = rma.mv(z,var.z, mod = ~ sqrt(var.z), random =  ~ 1 | author/cor_id, data = excitability_r,
test.egger
```

```
##
## Multivariate Meta-Analysis Model (k = 27; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0000  0.0000     18     no         author
## sigma^2.2  0.0000  0.0000     27     no  author/cor_id
##
## Test for Residual Heterogeneity:
## QE(df = 25) = 25.9966, p-val = 0.4078
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 25) = 0.0861, p-val = 0.7716
##
## Model Results:
##
##              estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt        0.0441  0.2415   0.1826  25  0.8566  -0.4534  0.5416
## sqrt(var.z)   -0.3570  1.2166  -0.2934  25  0.7716  -2.8627  2.1487
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
test.egger = rma.mv(z,var.z, mod = ~ sqrt(var.z), random =  ~ 1 | author/cor_id, data = inhibition_r, te
test.egger
```

```
##
## Multivariate Meta-Analysis Model (k = 20; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0222  0.1491     15     no         author
## sigma^2.2  0.0000  0.0000     20     no  author/cor_id
##
## Test for Residual Heterogeneity:
## QE(df = 18) = 27.4910, p-val = 0.0702
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 18) = 0.1178, p-val = 0.7354
##
## Model Results:
##
##              estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt       -0.1458  0.4286  -0.3401  18  0.7377  -1.0463  0.7547
## sqrt(var.z)    0.7925  2.3093   0.3432  18  0.7354  -4.0592  5.6442
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
test.egger = rma.mv(z,var.z, mod = ~ sqrt(var.z), random =  ~ 1 | author/cor_id, data = facilitation_r,
test.egger
```

```
##
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0000  0.0000      8     no         author
## sigma^2.2  0.0000  0.0000     14     no  author/cor_id
##
## Test for Residual Heterogeneity:
## QE(df = 12) = 10.3818, p-val = 0.5825
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 12) = 2.0249, p-val = 0.1802
##
## Model Results:
##
##              estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt        0.6294  0.4035   1.5598  12  0.1448  -0.2498  1.5085
## sqrt(var.z)   -2.9517  2.0743  -1.4230  12  0.1802  -7.4712  1.5678
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#Pulication Bias ADD TEXT

References

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